

Solutions to MP363 Problem Set 3

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P.1

The point here is to use the actual form of the Hamiltonian as a differential operator, i.e.

$$H = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(t, x)$$

to show $\langle \psi | \psi \rangle$ is time-independent, rather than use a more abstract approach. As hinted, this involves some calculations very close to Problem 4 in PS2; in particular, recall the identity

$$f \frac{\partial^2 g}{\partial x^2} - \frac{\partial^2 f}{\partial x^2} g = \frac{\partial}{\partial x} \left(f \frac{\partial g}{\partial x} - \frac{\partial f}{\partial x} g \right).$$

This will make another appearance later.

So let's go: since x is being integrated over and not t , the time derivative can be brought inside the integral:

$$\begin{aligned} \frac{\partial}{\partial t} \int_{-\infty}^{\infty} |\psi(t, x)|^2 dx &= \int_{-\infty}^{\infty} \frac{\partial}{\partial t} |\psi(t, x)|^2 dx \\ &= \int_{-\infty}^{\infty} \frac{\partial}{\partial t} (\psi^*(t, x) \psi(t, x)) dx \\ &= \int_{-\infty}^{\infty} \left[\frac{\partial \psi^*}{\partial t} \psi(t, x) + \psi^*(t, x) \frac{\partial \psi}{\partial t} \right] dx. \end{aligned}$$

Now, we have an expression for $\frac{\partial \psi}{\partial t}$: simply multiply Schrödinger's eqn through by $-i/\hbar$ to get

$$\frac{\partial \psi}{\partial t} = \frac{i\hbar}{2m} \frac{\partial^2 \psi}{\partial x^2} - \frac{i}{\hbar} V \psi$$

But complex conjugating both sides gives us $\frac{\partial \psi^*}{\partial t}$ as well (since t is a real number):

$$\frac{\partial \psi^*}{\partial t} = \left(\frac{i\hbar}{2m} \frac{\partial^2 \psi}{\partial x^2} - \frac{i}{\hbar} V \psi \right)^* = -\frac{i\hbar}{2m} \frac{\partial^2 \psi^*}{\partial x^2} + \frac{i}{\hbar} V \psi^*$$

since $\hbar, m, x$ and V are all real numbers. Thus, using these in the above integral gives

$$\begin{aligned} \frac{\partial}{\partial t} \int_{-\infty}^{\infty} |\psi(t, x)|^2 dx &= \int_{-\infty}^{\infty} \left\{ \left(-\frac{i\hbar}{2m} \frac{\partial^2 \psi^*}{\partial x^2} + \frac{i}{\hbar} V \psi^* \right) \psi + \psi^* \left(\frac{i\hbar}{2m} \frac{\partial^2 \psi}{\partial x^2} - \frac{i}{\hbar} V \psi \right) \right\} dx \\ &= \int_{-\infty}^{\infty} \left(-\frac{i\hbar}{2m} \frac{\partial^2 \psi^*}{\partial x^2} \psi + \frac{i\hbar}{2m} \psi^* \frac{\partial^2 \psi}{\partial x^2} + \frac{i}{\hbar} V \psi^* \psi - \frac{i}{\hbar} V \psi^* \psi \right) dx \\ &= \frac{i\hbar}{2m} \int_{-\infty}^{\infty} \left(\psi^* \frac{\partial^2 \psi}{\partial x^2} - \frac{\partial^2 \psi^*}{\partial x^2} \psi \right) dx \\ &= \frac{i\hbar}{2m} \int_{-\infty}^{\infty} \frac{\partial}{\partial x} \left(\psi^* \frac{\partial \psi}{\partial x} - \frac{\partial \psi^*}{\partial x} \psi \right) dx, \end{aligned}$$

using the identity we proved previously. But the fundamental theorem of

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calculus says that a definite integral of a total derivative is just the difference of the function at the endpoints, i.e.

$$\int_{-\infty}^{\infty} \frac{d}{dx} \left(\psi^* \frac{\partial \psi}{\partial x} - \frac{\partial \psi^*}{\partial x} \psi \right) dx = \left[\psi^* \frac{\partial \psi}{\partial x} - \frac{\partial \psi^*}{\partial x} \psi \right]_{-\infty}^{\infty} = 0$$

because $\psi(t, x)$ will vanish for both $x \rightarrow \infty$ and $x \rightarrow -\infty$ if $|\psi|^2$ has a finite integral. Thus, we have just proved

$$\frac{d}{dt} \int_{-\infty}^{\infty} |\psi(t, x)|^2 dx = 0$$

and thus $\langle \psi | \psi \rangle$ is time-independent. Thus, if it's equal to 1 at any particular time, it's equal to 1 for all time. Thus, $\langle \psi | \psi \rangle = 1$ can be imposed consistently.

P.2

This is just a matter of computing $\langle \psi | \psi \rangle$ (or $\langle u | u \rangle$) using the appropriate inner product, then setting to 1 and solving for the given constant. The point of this exercise is not just to show how we normalize states, but also how we work with inner products and states in different representations.

(a) Here, our states are functions and our inner products are integrals. So if $\psi = \alpha_1 x e^{-x}$, then

$$\langle \psi | \psi \rangle = \int_{-\infty}^{\infty} |\alpha_1 x e^{-x}|^2 dx = \alpha_1^2 \int_0^{\infty} x^2 e^{-2x} dx$$

(since we assume α_1 is real, $|\alpha_1|^2 = \alpha_1^2$.) The integral is easily

done if you integrate by parts twice, with the result being

$$\int_0^{\infty} x^2 e^{-2x} dx = \left[-\frac{1}{2} x^2 e^{-2x} - \frac{1}{2} x e^{-2x} - \frac{1}{4} e^{-2x} \right]_0^{\infty} = 1/4$$

(Recall $\lim_{x \rightarrow \infty} x^n e^{-x} = 0$ for any $n > 0$. Also, there's a formula for

the integral:

$$\int_0^{\infty} x^n e^{-ax} dx = \frac{n!}{a^{n+1}}$$

for any positive a and $n = 0, 1, 2, 3, \dots$ Just in case you ever need it.)

So we get

$$\langle \psi | \psi \rangle = \frac{\alpha_1^2}{4}$$

and thus, the real positive value that gives $\langle \psi | \psi \rangle = 1$ is $\boxed{\alpha_1 = 2}$.

(b) Here we have states given by 2-dimensional column vectors. The inner

product in this case is $\langle u | v \rangle = \sum_{i=1}^2 u_i^* v_i = u^\dagger \cdot v$

where † is the conjugate - transpose of the column matrix u . Note

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works,

$$\begin{aligned} \langle u|u \rangle &= \begin{vmatrix} -\alpha_2 \\ i\alpha_2 \end{vmatrix}^\dagger \begin{vmatrix} -\alpha_2 \\ i\alpha_2 \end{vmatrix} = \left[\begin{pmatrix} -\alpha_2 \\ i\alpha_2 \end{pmatrix}^\dagger \right]^* \begin{pmatrix} -\alpha_2 \\ i\alpha_2 \end{pmatrix} = (-\alpha_2 \ i\alpha_2)^* \begin{pmatrix} -\alpha_2 \\ i\alpha_2 \end{pmatrix} \\ &= (-\alpha_2^* \ -i\alpha_2^*) \begin{pmatrix} -\alpha_2 \\ i\alpha_2 \end{pmatrix} \\ &= (-\alpha_2)^* (-\alpha_2) + (-i\alpha_2^*) (i\alpha_2) = 2|\alpha_2|^2 = 2\alpha_2^2 \end{aligned}$$

Since we have assumed α_2 is real. Therefore, the positive constant that gives u a unit norm is $\boxed{\alpha_2 = \frac{1}{\sqrt{2}}}$

(c) Here we use Dirac's bra-ket notation, so $\langle \psi | \psi \rangle$ is actually

$\langle \psi | \cdot | \psi \rangle = (| \psi \rangle)^\dagger | \psi \rangle$. Since the adjoint operation turns kets into bras (and vice versa) and complex-conjugates numbers,

$$\begin{aligned} \langle \psi | &= (| \psi \rangle)^\dagger = \left(\sum_{m=0}^{\infty} \frac{\alpha_3}{3^m} | \phi_m \rangle \right)^\dagger = \sum_{m=0}^{\infty} \left(\frac{\alpha_3}{3^m} \right)^* (| \phi_m \rangle)^\dagger \\ &= \sum_{m=0}^{\infty} \frac{\alpha_3}{3^m} \langle \phi_m | \end{aligned}$$

Thus,

$$\langle \psi | \psi \rangle = \sum_{m=0}^{\infty} \frac{\alpha_3}{3^m} \langle \phi_m | \sum_{n=0}^{\infty} \frac{\alpha_3}{3^n} | \phi_n \rangle = \sum_{m,n=0}^{\infty} \frac{\alpha_3^2}{3^{m+n}} \langle \phi_m | \phi_n \rangle$$

$$= \alpha_3^2 \sum_{m,n=0}^{\infty} \frac{\delta_{m,n}}{3^{m+n}} = \alpha_3^2 \sum_{n=0}^{\infty} \frac{1}{3^n} \left(\sum_{m=0}^{\infty} \frac{\delta_{m,n}}{3^m} \right)$$

For a given n , $\sum_{m=0}^{\infty} \frac{\delta_{m,n}}{3^m}$ has terms which are zero for all values of m which are not n , and $\frac{1}{3^n}$ for the single value where m is equal to n .

Thus, $\sum_{m=0}^{\infty} \frac{\delta_{m,n}}{3^m} = \frac{1}{3^n}$, and so

$$\langle \psi | \psi \rangle = \alpha_3^2 \sum_{n=0}^{\infty} \frac{1}{3^n} \cdot \frac{1}{3^n} = \alpha_3^2 \sum_{n=0}^{\infty} \frac{1}{9^n} = \alpha_3^2 \frac{1}{1 - 1/9} = \frac{9\alpha_3^2}{8}$$

where we have used the formula for the sum of a geometric series:

$$\sum_{n=0}^{\infty} r^n = 1 + r + r^2 + r^3 + \dots = \frac{1}{1-r}$$

where $|r| < 1$, here $r = 1/9$ (which is indeed less than one), so we get

$\langle \psi | \psi \rangle$. Thus, for this to be 1, we see $\boxed{\alpha_3 = \frac{2\sqrt{2}}{3}}$ is the correct value.

P.3

This builds on P2 by making us compute $A| \psi \rangle$ for each case, i.e. the A given acts on the appropriate state in P.2. How this works depends on the representation, because A will act on $| \psi \rangle$ differently in each case. For example, if $| \psi \rangle$ is a wavefunction, then A could be a derivative or a multiplication by a function; if $| \psi \rangle$ is

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column vector, the A will be a matrix of the same dimension. And so on.

(a) Our state is $\alpha_1 x e^{-x}$, so $P\psi$ is

$$P\psi = -i\hbar \frac{\partial}{\partial x} (\alpha_1 x e^{-x}) = -i\hbar \alpha_1 (e^{-x} - x e^{-x}).$$

$$\text{Thus, } \langle \psi | P | \psi \rangle = \int_0^\infty \psi^* (P\psi) dx$$

$$\begin{aligned} &= \int_0^\infty (\alpha_1 x e^{-x}) (-i\hbar \alpha_1 (e^{-x} - x e^{-x})) dx \\ &= -i\hbar \alpha_1^2 \int_0^\infty x e^{-2x} dx + i\hbar \alpha_1^2 \int_0^\infty x^2 e^{-2x} dx \end{aligned}$$

Using the handy integral I gave earlier,

$$\int_0^\infty x e^{-2x} dx = \frac{1!}{2^{1+1}} = \frac{1}{4}, \quad \int_0^\infty x^2 e^{-2x} dx = \frac{2!}{2^{2+1}} = \frac{1}{4}$$

so

$$\langle \psi | P | \psi \rangle = -i\hbar \alpha_1^2 \left(\frac{1}{4} \right) + i\hbar \alpha_1^2 \left(\frac{1}{4} \right) = \boxed{0}$$

(b) L_z is a 2×2 matrix, so, for action on the 2-dim column vector u

$$L_z u = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \cdot \alpha_2 \begin{pmatrix} 1 \\ -i \end{pmatrix} = \frac{\alpha_2 \hbar}{2} \begin{pmatrix} 1 \\ -i \end{pmatrix}.$$

Thus,

$$\begin{aligned} \langle u | L_z u \rangle &= u^\dagger \cdot (L_z u) = \begin{pmatrix} -\alpha_2 \\ i\alpha_2 \end{pmatrix}^\dagger \cdot \begin{pmatrix} -\alpha_2 \hbar/2 \\ -i\alpha_2 \hbar/2 \end{pmatrix} \\ &= (-\alpha_2 \quad -i\alpha_2) \begin{pmatrix} -\alpha_2 \hbar/2 \\ -i\alpha_2 \hbar/2 \end{pmatrix} \\ &= \alpha_2^2 \hbar/2 - \alpha_2^2 \hbar/2 = \boxed{0}. \end{aligned}$$

(c) Finally, an expectation value that is not zero. It acts on $|\phi_n\rangle$ as

$$\begin{aligned} \text{given, so } H|\psi\rangle &= \sum_{n=0}^\infty \frac{\alpha_3}{3^n} H|\phi_n\rangle = \sum_{n=0}^\infty \frac{\alpha_3}{3^n} 2^n E_0 |\phi_n\rangle \\ &= \alpha_3 E_0 \sum_{n=0}^\infty \left(\frac{2}{3}\right)^n |\phi_n\rangle \end{aligned}$$

Thus,

$$\begin{aligned} \langle \psi | H | \psi \rangle &= \left(\sum_{m=0}^\infty \frac{\alpha_3}{3^m} \langle \phi_m | \right) \left(\alpha_3 E_0 \sum_{n=0}^\infty \left(\frac{2}{3}\right)^n |\phi_n\rangle \right) \\ &= \alpha_3^2 E_0 \sum_{n=0}^\infty \left(\frac{2}{3}\right)^n \sum_{m=0}^\infty \frac{\langle \phi_m | \phi_n \rangle}{3^m} \\ &= \alpha_3^2 E_0 \sum_{n=0}^\infty \left(\frac{2}{3}\right)^n \sum_{m=0}^\infty \frac{\delta_{mn}}{3^m} = \alpha_3^2 E_0 \sum_{n=0}^\infty \left(\frac{2}{3}\right)^n \frac{1}{3^n} \\ &= \alpha_3^2 E_0 \sum_{n=0}^\infty \left(\frac{2}{3}\right)^n = \alpha_3^2 E_0 \frac{1}{1-2/3} = \boxed{\frac{9\alpha_3^2 E_0}{7}} \end{aligned}$$

or, using $\alpha_3 = 2\sqrt{2}/3$,

$$\boxed{\langle H \rangle = \frac{8}{7} E_0}$$

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P.4

(a) This is very straightforward using the defn of the commutator: the right-hand side is

$$\begin{aligned} A[B, C] + [A, C]B &= A(BC - CB) + (AC - CA)B \\ &= ABC - ACB + ACB - CAB \\ &= ABC - CAB = (AB)C - C(AB) \\ &= [AB, C] \end{aligned}$$

This identity can be quite useful. For instance, in the next part...

(b) Recall that we prove something by induction in the following way:

if S_n is the thing we want to check, we must first make sure that our statement is true for $n=1$, i.e. S_1 holds. Then, we want to show that S_n being true implies that S_{n+1} is also true. If we can do both these, then S_n is true for all n .

So we start by knowing $[X, P] = i\hbar 1$; now consider $[X^n, P] = i\hbar n X^{n-1}$. Is it true for $n=1$? $[X^1, P] = [X, P] = i\hbar 1$, and we have $i\hbar n X^{n-1} = i\hbar (1) X^0 = i\hbar 1$ for $n=1$, so $[X^n, P] = i\hbar n X^{n-1}$ holds for $n=1$.

Now assume that $[X^n, P] = i\hbar n X^{n-1}$ holds for a value of n , and we look at the left-hand side $[X^{n+1}, P]$. X^{n+1} can be written as $X^n \cdot X$, so $[X^{n+1}, P] = [X^n \cdot X, P]$. So let's $A = X^n$, $B = X$ and $C = P$ in the identity we proved in (a), we see

$[X^{n+1}, P] = X^n [X, P] + [X^n, P] X$. We know that $[X, P] = i\hbar 1$, and we have assumed that $[X^n, P] = i\hbar n X^{n-1}$. Thus,

$$\begin{aligned} [X^{n+1}, P] &= X^n (i\hbar 1) + (i\hbar n X^{n-1}) X \\ &= i\hbar X^n + i\hbar n X^n = i\hbar (n+1) X^n \end{aligned}$$

which is the relation we wished to prove. Thus, $[X^n, P] = i\hbar n X^{n-1}$

$\Rightarrow [X^{n+1}, P] = i\hbar (n+1) X^n$, and so $[X^n, P] = i\hbar n X^{n-1}$ for all

positive integers n .

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The proof that $[X, P^n] = i\hbar n P^{n-1}$ is identical; $n=1$ gives $[X, P] = i\hbar$, which is true. And

$$\begin{aligned}
 [X, P^{n+1}] &= [X, P^n P] = -[P^n, P, X] \\
 &= -(P^n [P, X] + [P^n, X] P) \\
 &= -(P^n (-i\hbar) - [X, P^n] P) \\
 &= i\hbar P^n + [X, P^n] P \\
 &= i\hbar P^n + (i\hbar n P^{n-1}) P \\
 &= i\hbar P^n + i\hbar n P^n \\
 &= i(n+1)\hbar P^n
 \end{aligned}$$

$$\text{and so } [X, P^n] = i\hbar n P^{n-1} \text{ for } n=1, 2, 3, \dots$$

(These two commutators will pop up from time to time; we'll see them again when we discuss the QM simple harmonic oscillator.)